Emily Dombrowski
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Education

University of Vermont 2023-Present
Ph.D. candidate in Biology

College of Charleston 2019-2023
B.S. Marine Biology
B.A. French and Francophone Studies

Research Funding and Awards

Chateaubriand Fellowship, Embassy of France (€16,432)	2026
Wheeler Award, University of Vermont ($2,000)	2026
Suiter Award, University of Vermont ($1,000)	2026
Graduate Research Fellowship, National Science Foundation ($159,000)	2025
Barry Goldwater Scholarship, Goldwater Foundation ($7,500)	2022
Hollings Scholarship, National Oceanic and Atmospheric Administration ($26,000)	2021
Research Experience for Undergraduates, National Science Foundation ($7,200)	2021
Summer Undergraduate Research Funding, College of Charleston ($4,000)	2020
Charleston Fellows Program and Swanson Family Scholarship	2019-2023

Publications

Crane, L. C., **Dombrowski, E**., Gutzler, B. C., Jury, S. H., Bradt, G., & Goldstein, J. S. (2025). Hemolymph-based molt prediction in the European green crab, *Carcinus maenas*: Potential tools for a Soft-Shell industry. *Journal of Shellfish Research*, 44(1), 153–161.

Research Experience

Larval selection and temperature tolerance in a marine mussel hybrid zone 2023-present
Advisors: Dr. Brent Lockwood, University of Vermont, USA; Dr. Cynthia Riginos, University of Queensland, Australia; Dr. Bastien Saint-Leandre, University Le Havre, France

- PhD dissertation evaluating replicated *Mytilus* mussel hybrid zones in Australia and France in the context of changing temperatures
- Designed and completed larval selection experiments in Australia (2025) to understand how a hybrid zone along a thermal gradient genomically responds to temperature
- Obtaining funding to complete work in France for transgenerational plasticity experiments (2026-2027)

- Following data collection, international systems will be synthesized into an evolutionary simulation

Using Hemolymph Chemistry to Predict Molting in Green Crabs, ***Carcinus maenas*** 2022
Advisor: Dr. Jason Goldstein, Wells National Estuarine Research Reserve

- Conducted hemolymph draws and chemistry analyses, ran correlational statistics in R for NOAA Hollings Scholarship summer internship
- Prepared final presentation to be presented with NOAA and National Estuarine Research Reserves (NERR)

Horseshoe crab, *Limulus* polyphemus, temperature tolerance and physiology 2021-2023
Advisor: Dr. Jody M. Beers

- Produced novel heart rate imaging technique for horseshoe crabs following commercial hemolymph draw, prepared research procedures, oversaw animal husbandry, scientific communication training
- Research used as pilot study for South Carolina Sea Grant Project awarded to Beers Lab (2021)

Physiological effects of temperature and parasite load on spotted seatrout, *Cynoscion nebulosus* 2020 to 2021
Advisor: Dr. Jody M. Beers

- Conducted blood and enzymatic analyses, water parameter testing, and data analysis

Presentations

Dombrowski, E., Nunez J., Lockwood, B. Using a quantitative trait nucleotide simulation to assess the impact of migration and thermal selection on local adaptation in a mussel hybrid zone. 3rd Joint Congress on Evolutionary Biology. 29 July 2024. (Poster)

Dombrowski, E., Sasson, D., Beers, J. Physiological Effects of Age in Atlantic Horseshoe Crabs, *Limulus Polyphemus.* Benthic Ecology Meeting. 27 April 2023. (Poster)

Dombrowski, E., Crane, L., Jury, S., Gutzler, B., Bradt, G., Goldstein, J. Using Hemolymph Chemistry to Predict and Assess Molting in Green Crabs, *Carcinus maenas*.

- Regional Association for Research on the Gulf of Maine (RARGOM). 2 December 2022. (Talk)
- College of Charleston School of Sciences, Mathematics, and Engineering 2022 Research Poster Session. 22 August 2022. (Poster)
- Hollings and Educational Partnership Program with Minority Serving Institutions Scholars and the National Estuarine Research Reserves Research Symposium. 28 July 2022. (Talk)

Dombrowski, E., Beers J. Temperature Tolerance and Physiology of the Horseshoe Crab, *Limulus polyphemus*.

- College of Charleston School of Sciences, Mathematics, and Engineering 2022 Research Poster Session. 7 April 2022. (Poster)
- Resilience and Response of Marine Organisms, South Carolina Department of Natural Resources. 4 August 2021. (Talk)

Dombrowski, E., Beers J. Physiological effects of temperature and parasite load on spotted seatrout, *Cynoscion nebulosus.* School of Science and Math Poster Session, College of Charleston. April 2021. (Video)

Teaching and Mentoring Experience

Graduate writing tutor 2026-Present

- Peer mentor to graduate students working on writing assignments across disciplines
- Designing and presenting workshops for graduate writers about overcoming writing challenges and strengthening work

Teaching assistant:

Genetics Recitation	Spring 2025
Comparative Physiology Lab	Fall 2024
Principles of Biology II Lab	Spring 2024
Principles of Biology I Lab	Fall 2023

Guest lectures:

- *Getting Unstuck*. Proposal Writing class, University of Vermont, Department of Biology, March 2026.
- *Personal Statements for Health Professions.* Workshop, University of Vermont, Post Baccalaureate and Pre-Medical and Medical Science Departments, February 2026.
- *RNA Transcription*. Genetics lecture, University of Vermont, Department of Biology, February 2025.

Community Outreach

Graduate Firsts Club Founder, University of Vermont 2025-Present

- Founded and obtained funding for a club that creates community and mentorship opportunities for First Generation Graduate Students
- Established ongoing collaborations between undergraduate Firsts Club and Society for Advancement of Chicanos/Hispanics & Native Americans in Science (SACNAS) organizations to facilitate career-oriented programming
- Hosts monthly coffee networking events and career focused-work shops

TriBeta Biology Honors Society Professional Development Series, University of Vermont 2025-Present

- Developed a series of professional development workshops for undergraduate students in the TriBeta Honors society to share resources about applying to graduate school, creating a resume, and writing cover letters

- Presented resources, document edits, and techniques to over 60 students during a 3 part workshop series

Grant writing intern and committee member for the Lonon Foundation 2019-2023

- Authored grants that amounted to $16,000 for resource kits, free monthly programming, and mental health support for children whose caregivers have been diagnosed with cancer in the South Carolina Lowcountry region
- Positions: Hike for Mike Committee Member (2021-2022), Volunteer Engagement Chair (2020-2023)
- Responsibilities: Oversee volunteers, plan monthly events for children, plan annual events for families, communicate with sponsors, facilitate events

Science Communication Events

- In conjunction with The Wells National Estuarine Research Reserve, The Lonon Foundation, and my research, I have developed marine organism-oriented curriculum to present to local children to enhance community understanding and interaction with science
- Healthy Rivers Ogunquit Kayak Tour: Facilitated programming about invasive species in the Gulf of Maine to present to kayakers on the Ogunquit River
- Event facilitator: Minority Women in STEM Camp Session with Portland High School (June 2022), Mysteries Under the Ocean with Camp Rise Above (June 2022), Marine Biology Virtual Field Day with the Lonon Foundation (September 2021)
- Community Presenter: Cultivate Art and Science (July 2021) Planned and facilitated SciComm events to contribute to scientific discussion in Charleston community for children and adults

Academic Service

Peer reviewer, Scientific Reports 2025

Graduate Affairs Student Liaison, University of Vermont 2024-Present

- Communicates graduate student quality of life survey data between students and department faculty and leads graduate student driven initiatives

Laboratory Safety Officer, Lockwood Lab 2023-Present

Society Memberships

American Malacological Society 2026-Present
Society for Integrative and Comparative Biology 2025-Present